Rakamin Academy X kimia farma

# Business Performance Analytics 2020 - 2023

## Kimia Farma - Big Data Analytics

Presented by
Aliyya Najla Syaharani

Rakamin Academy

# Aliyya Najla Syaharani

## Aspiring Data Analyst & Nutrition Science Graduate

I am an aspiring Data Analyst passionate about transforming data into meaningful insights. Through my Fullstack Data Analysis bootcamp, I developed skills in statistics, problem solving, SQL, Python, and data visualization using Google Looker, with hands-on experience in data cleaning and analysis. My undergraduate background in Nutrition Science strengthened my analytical thinking, attention to detail, and ability to interpret quantitative data in a structured way. I am committed to continuously improving my skills to deliver impactful, data-driven solutions.

Bogor, Indonesia

aliyyanajla11@gmail.com

Aliyya Najla Syaharani

# Courses and Certification

- Data Analysis Fullstack Intensive Bootcamp by MySkill (January, 2026) <certificate link>
- Data Analytics Mini Course by RevoU (November, 2025) <certificate link>
- Certified Microsoft Excel Course by Kursus Digital (Maret, 2026) <certificate link>

# About Company

1. **Historical Foundation & Transformation**
   Established in 1817 as NV Chemicalien Handle Rathkamp & Co., Kimia Farma is the first pharmaceutical industrial company in Indonesia. In 1958, following nationalization, several pharmaceutical entities were consolidated into PNF Bhinneka Kimia Farma. On August 16, 1971, the company officially became PT Kimia Farma (Persero), marking its transition into a Limited Liability Company.

2. **Public & Integrated Healthcare Company**
   In 2001, Kimia Farma became a publicly listed company under the name PT Kimia Farma (Persero) Tbk. Today, it operates as an integrated healthcare company, spanning pharmaceutical manufacturing, distribution, pharmacy retail, and healthcare services, supported by more than two centuries of industry experience.

# Project Overview

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## BACKGROUND

- Kimia Farma operates multiple branches across Indonesia and generates large volumes of transaction, product, inventory, and branch data.
- To evaluate business performance from 2020–2023, these datasets must be integrated and transformed into a structured analytical table using BigQuery.
- This project aims to build an interactive dashboard in Looker Studio to support data-driven decision making.

## PROBLEM STATEMENT

- Kimia Farma's data is stored in separate datasets (transactions, products, inventory, branches), making it difficult to:
  a. Monitor total transactions, sales, and profit
  b. Evaluate branch and provincial performance
  c. Analyze customer ratings
  d. Assess pricing and discount effectiveness
- Without a consolidated analytics table and dashboard, performance evaluation and strategic planning become less effective.

Project Explanation Video Here

GitHub Link Here

# Project Overview

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## BUSINESS QUESTIONS

1. How did Kimia Farma perform overall from 2020 to 2023 in terms of total transactions, nett sales, nett profit, and customer transaction ratings?
2. What sales trends can be observed year-over-year, and which year recorded the strongest profitability?
3. Which provinces contribute the highest transaction volumes and nett sales, and does transaction activity align with profit generation?
4. How is profitability distributed geographically across Indonesia, and are there regional performance gaps?
5. Which branch categories generate the greatest revenue contribution, and do certain formats outperform others?
6. Are there discrepancies between branch ratings and transaction ratings that may indicate service or operational issues?
7. How effective is the current pricing, discount, and gross profit structure in supporting sustainable profitability, and which areas require strategic improvement?

# Data Overview

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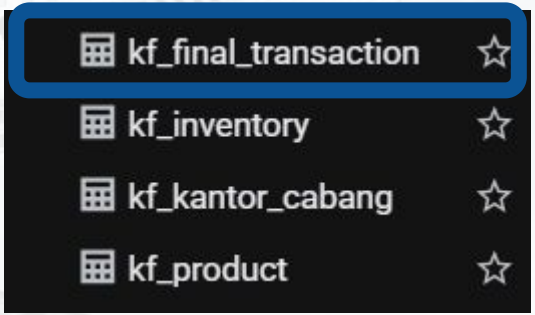
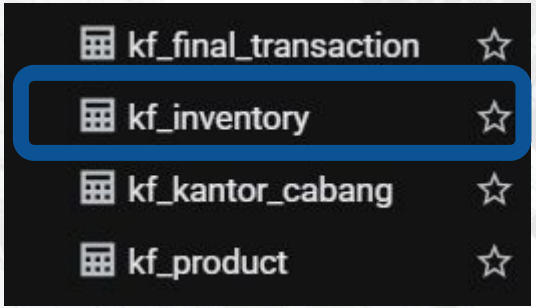



**kf_final_transaction**
Contains detailed transaction records, including transaction_id, transaction date, branch_id, customer_name, product_id, product price, discount_percentage, and transaction rating.

**kf_inventory**
Provides stock information for each branch, including inventory_id, branch_id, product_id, product_name, and opname_stock.



# Data Overview

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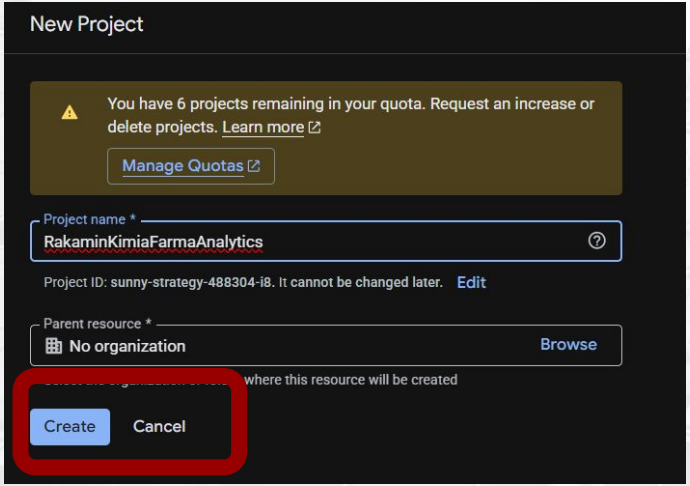
- kf_final_transaction
- kf_inventory
- kf_kantor_cabang
- kf_product

**kf_kantor_cabang**
Contains branch-level information such as branch_id, branch_category, branch_name, city (kota), province (provinsi), and branch rating.

**kf_product**
Lists product details, including product_id, product_name, product_category, and product price.

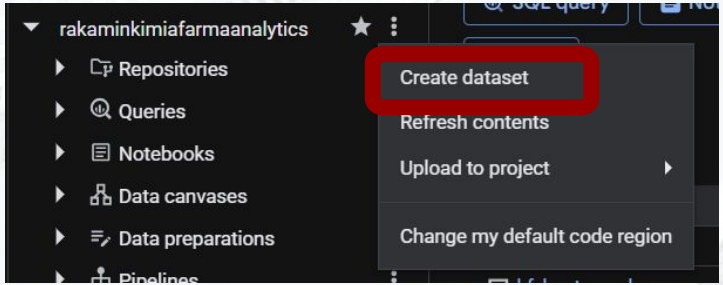

- kf_final_transaction
- kf_inventory
- kf_kantor_cabang
- kf_product

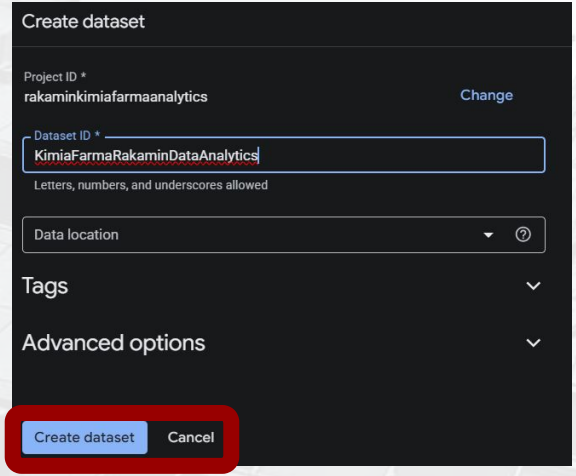
# Create a New Project in BigQuery




1. Go to **Google Cloud Console** & log in using Google account
2. At the top of the page, click **"Select a Project"**
3. Click **"New Project"**
4. Enter the following project name: RakaminKimiaFarmaAnalytics
5. Click **"Create"**

# Create a Dataset in BigQuery




1. Navigate to **BigQuery** from the left navigation menu.
2. Make sure you are inside the project: RakaminKimiaFarmaAnalytics
3. Click **"Create Dataset."**
4. Enter the Dataset ID as: KimiaFarmaRakaminDataAnalytics
5. Click **"Create Dataset"**

# Create a Table in BigQuery

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1. Click **“Create Table”**
2. Under **Source**:
   a. Create table from → Upload
   b. Select the CSV file (e.g., kf_final_transaction.csv)
   c. File format → CSV
3. Under **Destination**:
   a. Project → rakaminkimiafarmaanalytics
   b. Dataset → KimiaFarmaRakaminDataAnalytics
   c. Table name → Use the file name without .csv
      Example: kf_final_transaction
4. Click “**Create Table**”

# Create a Table in BigQuery

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Create table

**Source**

Upload

Select file *
kf_final_transaction.csv
Browse

File format
CSV

**Destination**

rakaminkimiafarmaanalytics
Browse

Dataset *
KimiaFarmaRakaminDataAnalytics

Table *
kf_final_transaction

Maximum name size is 1,024 UTF-8 bytes. Unicode letters, marks, numbers, connectors, dashes, and spaces are allowed.

Table type
Native table

**Schema**

Create table
Cancel

Search BigQuery resources
Show starred only
rakaminkimiafarmaanalytics
Repositories
Queries
Notebooks
Data canvases
Data preparations
Pipelines
Connections
KimiaFarmaRakaminData
kf_final_

Open
Open in
Create table
Create ML model
Share
Copy ID
Refresh contents
Delete

# Data Preparation

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cleaned_trans... Query Create convers

Schema Details Preview Table Explorer Preview

Filter Enter property name or value

| Field name | Type | Mode |
|---|---|---|
| transaction_id | STRING | NULLABLE |
| date | DATE | NULLABLE |
| branch_id | INTEGER | NULLABLE |
| customer_name | STRING | NULLABLE |
| product_id | STRING | NULLABLE |
| price | FLOAT | NULLABLE |
| discount_percentage | FLOAT | NULLABLE |
| rating | FLOAT | NULLABLE |

**Cleaned Transactions Table**

```sql
CREATE OR REPLACE TABLE `rakaminkimiafarmaanalytics.KimiaFarmaRakaminDataAnalytics.cleaned_transactions` AS
SELECT
    transaction_id,
    DATE(date) AS date,
    branch_id,
    customer_name,
    product_id,
    SAFE_CAST(price AS FLOAT64) AS price,
    SAFE_CAST(discount_percentage AS FLOAT64) AS discount_percentage,
    SAFE_CAST(rating AS FLOAT64) AS rating
FROM `rakaminkimiafarmaanalytics.KimiaFarmaRakaminDataAnalytics.kf_final_transaction`
WHERE price IS NOT NULL
  AND discount_percentage IS NOT NULL;
```

# Data Preparation

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## Cleaned Inventory Table

cleaned_inven... Query Create conversatio

Schema Details Preview Table Explorer Preview

Filter Enter property name or value

| Field name | Type | Mode | Descripti |
|---|---|---|---|
| inventory_id | STRING | NULLABLE | - |
| branch_id | INTEGER | NULLABLE | - |
| product_id | STRING | NULLABLE | - |
| product_name | STRING | NULLABLE | - |
| opname_stock | INTEGER | NULLABLE | - |

```sql
CREATE OR REPLACE TABLE `rakaminkimiafarmaanalytics.KimiaFarmaRakaminDataAnalytics.cleaned_inventory` AS
SELECT
    inventory_id,
    branch_id,
    product_id,
    product_name,
    SAFE_CAST(opname_stock AS INT64) AS opname_stock
FROM `rakaminkimiafarmaanalytics.KimiaFarmaRakaminDataAnalytics.kf_inventory`
WHERE opname_stock IS NOT NULL;
```

# Data Preparation

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cleaned_bran... Query Create conve

Schema Details Preview Table Explorer Preview

Generate table and column descriptions with Gemini

Filter Enter property name or value

| Field name | Type | Mode |
|---|---|---|
| branch_id | INTEGER | NULLABLE |
| branch_name | STRING | NULLABLE |
| kota | STRING | NULLABLE |
| provinsi | STRING | NULLABLE |
| branch_category | STRING | NULLABLE |
| rating | FLOAT | NULLABLE |

**Cleaned Branch (*Kantor Cabang*) Table**

```google_sql_query
CREATE OR REPLACE TABLE `rakaminkimiafarmaanalytics.KimiaFarmaRakaminDataAnalytics.cleaned_branches` AS
SELECT
    branch_id,
    branch_name,
    kota,
    provinsi,
    branch_category,
    SAFE_CAST(rating AS FLOAT64) AS rating
FROM `rakaminkimiafarmaanalytics.KimiaFarmaRakaminDataAnalytics.kf_kantor_cabang`
```

# Data Preparation

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## Cleaned Product Table

cleaned_prod... Query Create cor

Schema Details Preview Table Explorer Previe

Generate table and column descriptions with Gem

Filter Enter property name or value

| Field name | Type | Mode |
|---|---|---|
| product_id | STRING | NULLABLE |
| product_name | STRING | NULLABLE |
| product_category | STRING | NULLABLE |
| price | FLOAT | NULLABLE |

```sql
CREATE OR REPLACE TABLE `rakaminkimiafarmaanalytics.KimiaFarmaRakaminDataAnalytics.cleaned_products` AS
SELECT
    product_id,
    product_name,
    product_category,
    SAFE_CAST(price AS FLOAT64) AS price
FROM `rakaminkimiafarmaanalytics.KimiaFarmaRakaminDataAnalytics.kf_product`
WHERE price IS NOT NULL;
```

# Building the Final Analysis Table

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## TABLE CREATION

The query uses **CREATE OR REPLACE TABLE** to generate the final analytical table named **kf_table_analysis**.

## DATA PREPARATION WITH CTE (WITH BASE AS)

A Common Table Expression (CTE) named **base** is created to organize the joined data before performing final calculations → improves readability and avoids repeating complex logic.

## TABLE INTEGRATION (LEFT JOIN)

- **LEFT JOIN** ensures all transaction records are retained, even if some product or branch information is missing
- **cleaned_transactions (t)** → main transaction dataset.
- **cleaned_products (p)** is joined using product_id to retrieve product names.
- **cleaned_branches (b)** is joined using branch_id to retrieve branch and location details.

# Building the Final Analysis Table

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## DATA TRANSFORMATION

- The **price** column is renamed to **actual_price** for clarity.
- Gross profit percentage is calculated using a tier-based pricing rule:
  - ≤ 50,000 → 10%
  - 50k–100k → 15%
  - 100k–300k → 20%
  - 300k–500k → 25%
  - 500k → 30%

## PROFITABILITY CALCULATIONS

- These calculated metrics enable profitability and performance analysis.
- Nett Sales = Actual Price – (Actual Price × Discount Percentage)
- Nett Profit = Nett Sales × Gross Profit Percentage

# Building the Final Analysis Table

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kfTableAnalysis | Run | Save query | Download | Share

```google_sql_query
CREATE OR REPLACE TABLE
`rakaminkimiafarmaanalytics.KimiaFarmaRakaminDataAnalytics.kf_table_analysis` AS

WITH base AS (
    SELECT
        t.transaction_id,
        t.date,
        t.branch_id,
        b.branch_name,
        b.kota,
        b.provinsi,
        b.rating AS rating_cabang,
        t.customer_name,
        t.product_id,
        p.product_name,
        t.price AS actual_price,
        t.discount_percentage,
        t.rating AS rating_transaksi,

        CASE
            WHEN t.price <= 50000 THEN 0.10
            WHEN t.price <= 100000 THEN 0.15
            WHEN t.price <= 300000 THEN 0.20
            WHEN t.price <= 500000 THEN 0.25
            ELSE 0.30
        END AS persentase_gross_laba

    FROM
    `rakaminkimiafarmaanalytics.KimiaFarmaRakaminDataAnalytics.cleaned_transactions` t

    LEFT JOIN
    `rakaminkimiafarmaanalytics.KimiaFarmaRakaminDataAnalytics.cleaned_products` p
    ON t.product_id = p.product_id

    LEFT JOIN
    `rakaminkimiafarmaanalytics.KimiaFarmaRakaminDataAnalytics.cleaned_branches` b
    ON t.branch_id = b.branch_id
)

SELECT *,
    (actual_price - (actual_price * discount_percentage)) AS nett_sales,
    (actual_price - (actual_price * discount_percentage))
        * persentase_gross_laba AS nett_profit
FROM base;
```

## Join All Tables in One Analysis Table

| Field name | Type | Mode | Desc |
|---|---|---|---|
| transaction_id | STRING | NULLABLE | - |
| date | DATE | NULLABLE | - |
| branch_id | INTEGER | NULLABLE | - |
| branch_name | STRING | NULLABLE | - |
| kota | STRING | NULLABLE | - |
| provinsi | STRING | NULLABLE | - |
| rating_cabang | FLOAT | NULLABLE | - |
| customer_name | STRING | NULLABLE | - |
| product_id | STRING | NULLABLE | - |
| product_name | STRING | NULLABLE | - |
| actual_price | FLOAT | NULLABLE | - |
| discount_percentage | FLOAT | NULLABLE | - |
| rating_transaksi | FLOAT | NULLABLE | - |
| persentase_gross_laba | FLOAT | NULLABLE | - |
| nett_sales | FLOAT | NULLABLE | - |
| nett_profit | FLOAT | NULLABLE | - |

# Data Insight

| Total Transaction | Total Sales | Total Profit | Transaction Rating |
|---|---|---|---|
| 672.5K | Rp321.17B | Rp91.21B | 4.0 |

# Data Analysis

## Yearly Sales Comparison

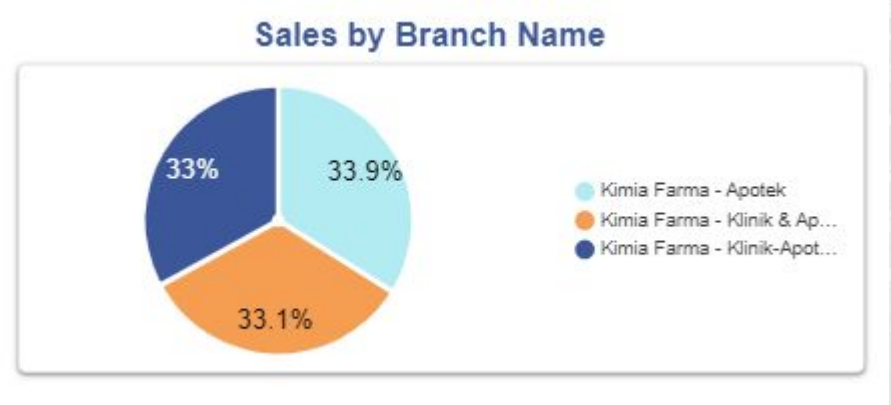
| Row | year | nett_sales |
|---|---|---|
| 1 | 2020 | 80437605040.0 |
| 2 | 2021 | 80037846824.0 |
| 3 | 2022 | 80578445844.0 |
| 4 | 2023 | 80117292611.0 |

```sql
SELECT
    EXTRACT(YEAR FROM date) AS year,
    ROUND(SUM(nett_sales), 2) AS nett_sales
FROM
`rakaminkimiafarmaanalytics.KimiaFarmaRakaminDataAnalytics.kf_table_analysis`
GROUP BY year
ORDER BY year;
```

# Data Analysis

## Sales by Branch Category

| Row | branch_category | total_sales |
|---|---|---|
| 1 | Apotek | 108889166820.0 |
| 2 | Klinik & Apotek | 106320084213.0 |
| 3 | Klinik-Apotek-Laboratorium | 105961939286.0 |



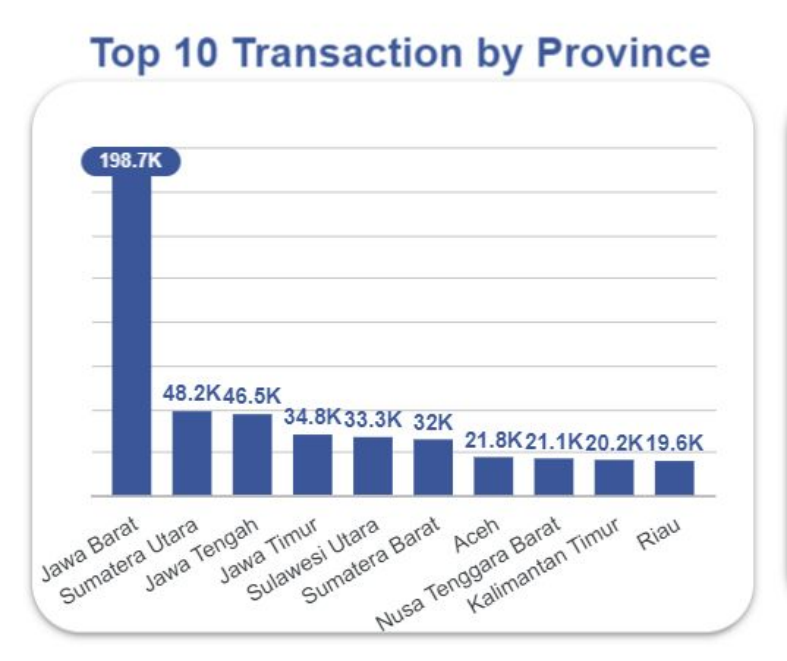

```sql
SELECT
    b.branch_category,
    SUM(a.nett_sales) AS total_sales
FROM
`rakaminkimiafarmaanalytics.KimiaFarmaRakaminDataAnalytics.kf_table_analysis` a
JOIN
`rakaminkimiafarmaanalytics.KimiaFarmaRakaminDataAnalytics.cleaned_branches` b
ON a.branch_id = b.branch_id
GROUP BY b.branch_category
ORDER BY total_sales DESC;
```

# Data Analysis

## Top 10 Total Transaction by Province

| Row | provinsi | total_transaction |
|---|---|---|
| 1 | Jawa Barat | 198723 |
| 2 | Sumatera Utara | 48178 |
| 3 | Jawa Tengah | 46494 |
| 4 | Jawa Timur | 34766 |
| 5 | Sulawesi Utara | 33339 |
| 6 | Sumatera Barat | 32014 |
| 7 | Aceh | 21825 |
| 8 | Nusa Tenggara Barat | 21069 |
| 9 | Kalimantan Timur | 20243 |
| 10 | Riau | 19607 |

```
SELECT
    provinsi,
    COUNT(transaction_id) AS total_transaction
FROM
`rakaminkimiafarmaanalytics.KimiaFarmaRakaminDataAnalytics.kf_table_analysis`
GROUP BY provinsi
ORDER BY total_transaction DESC
LIMIT 10;
```

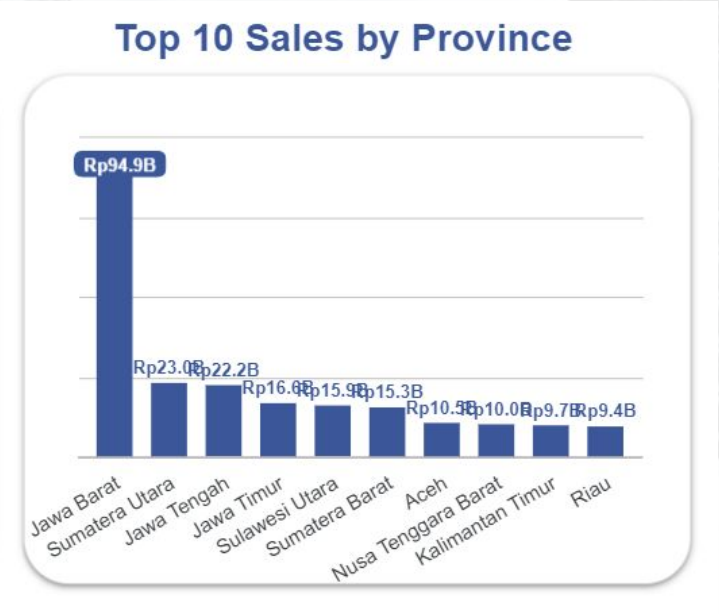


# Data Analysis

## Top 10 Total Sales by Province

| Row | provinsi | total_sales |
|---|---|---|
| 1 | Jawa Barat | 94869594875.0 |
| 2 | Sumatera Utara | 22952159897.0 |
| 3 | Jawa Tengah | 22248101144.0 |
| 4 | Jawa Timur | 16627080704.0 |
| 5 | Sulawesi Utara | 15902759535.0 |
| 6 | Sumatera Barat | 15314406695.0 |
| 7 | Aceh | 10467971437.0 |
| 8 | Nusa Tenggara Barat | 10036738806.0 |
| 9 | Kalimantan Timur | 9714536616.0 |
| 10 | Riau | 9354616403.0 |

```sql
SELECT
    provinsi,
    SUM(nett_sales) AS total_sales
FROM
`rakaminkimiafarmaanalytics.KimiaFarmaRakaminDataAnalytics.kf_table_analysis`
GROUP BY provinsi
ORDER BY total_sales DESC
LIMIT 10;
```



# Data Analysis

## Top 5: High Branch Rating, Low Transaction Rating

```sql
SELECT
    kota,
    branch_name AS branch,
    ROUND(AVG(rating_cabang), 2) AS branch_rating,
    ROUND(AVG(rating_transaksi), 2) AS transaction_rating
FROM
`rakaminkimiafarmaanalytics.KimiaFarmaRakaminDataAnalytics.kf_table_analysis`
GROUP BY kota, branch_name
HAVING AVG(rating_cabang) >= 4.5
   AND AVG(rating_transaksi) <= 4.0
ORDER BY branch_rating DESC
LIMIT 5;
```

| Row | kota | branch | branch_rating | transaction_rating |
|---|---|---|---|---|
| 1 | Pematangsiantar | Kimia Farma - Klinik-Apotek-Laboratorium | 4.83 | 3.99 |
| 2 | Batam | Kimia Farma - Klinik-Apotek-Laboratorium | 4.73 | 4.0 |
| 3 | Sorong | Kimia Farma - Apotek | 4.73 | 3.99 |
| 4 | Cilacap | Kimia Farma - Klinik-Apotek-Laboratorium | 4.71 | 4.0 |
| 5 | Solok | Kimia Farma - Klinik & Apotek | 4.7 | 3.99 |

### Top 5 High Branch Rating but Low Transaction Rating

| | City | Branch Name | Branch Rating ❶ | Transaction Rating ❷ |
|---|---|---|---|---|
| 1. | Solok | Kimia Farma - Klinik-Apotek-Laboratorium | 5 | 3.46 |
| 2. | Lubuklinggau | Kimia Farma - Klinik-Apotek-Laboratorium | 5 | 3.53 |
| 3. | Tebing Tinggi | Kimia Farma - Klinik & Apotek | 4.95 | 3.51 |
| 4. | Jambi | Kimia Farma - Klinik & Apotek | 4.94 | 3.5 |
| 5. | Kupang | Kimia Farma - Klinik-Apotek-Laboratorium | 4.93 | 3.51 |

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# Data Analysis

## Profit by Province

```
SELECT
    provinsi,
    SUM(nett_profit) AS total_profit
FROM
`rakaminkimiafarmaanalytics.KimiaFarmaRakaminDataAnalytics.kf_table_analysis`
GROUP BY provinsi
ORDER BY total_profit DESC;
```

| Row | provinsi | total_profit |
|---|---|---|
| 1 | Jawa Barat | 26940703195.150444 |
| 2 | Sumatera Utara | 6517187359.84993 |
| 3 | Jawa Tengah | 6318005348.2499857 |
| 4 | Jawa Timur | 4722764504.1999769 |
| 5 | Sulawesi Utara | 4516325323.5999966 |
| 6 | Sumatera Barat | 4349184029.4500208 |
| 7 | Aceh | 2973594696.1000018 |
| 8 | Nusa Tenggara Barat | 2848908857.5999956 |
| 9 | Kalimantan Timur | 2761089143.3500028 |
| 10 | Riau | 2656281862.1999989 |
| 11 | Kepulauan Riau | 2161206400.6999974 |
| 12 | Sumatera Selatan | 1972617932.1000051 |
| 13 | Bali | 1818321849.6500056 |

# KIMIA FARMA BUSINESS PERFORMANCE ANALYSIS DASHBOARD (2020–2023)

kimia farma

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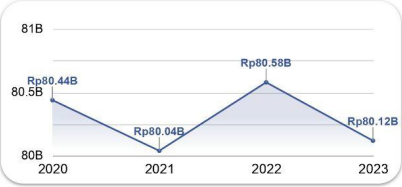
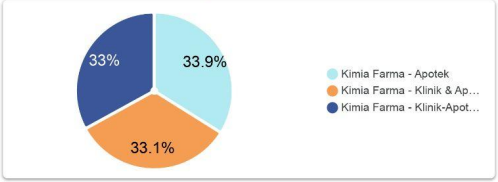
| Total Transaction | Total Sales | Total Profit | Transaction Rating |
|---|---|---|---|
| 672.5K | Rp321.17B | Rp91.21B | 4.0 |

**Dashboard Link**

## Yearly Sales Comparison




## Sales by Branch Name

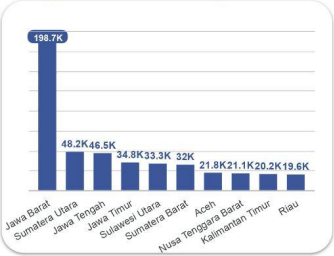



Province

Branch Name

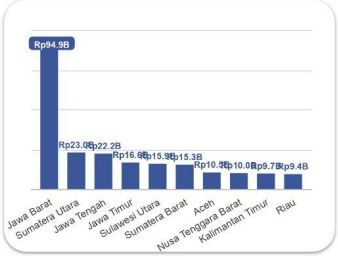
Select date range

## Top 10 Transaction by Province




## Total Profit by Province

## Top 10 Sales by Province




## Top 5 High Branch Rating but Low Transaction Rating

| | City | Branch Name | Branch Rating | Transaction Rating |
|---|---|---|---|---|
| 1. | Solok | Kimia Farma - Klinik-Apotek-Laboratorium | 5 | 3.46 |
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# Data Visualization Dashboard

# Summary

1. Revenue remained stable from 2020–2023, generating Rp321.17B with a **peak in 2022** and slight softening in 2023.
2. **West Java** leads performance, dominating both **transactions (198.7K)** and **sales (Rp94.9B)**, while contributions from **eastern regions remain relatively lower**.
3. Several branches show a service gap → **high branch ratings (~5.0)** but **lower transaction ratings (~3.5–4.0),** indicating operational improvement needs.
4. **Sales** is **evenly distributed** across branch formats, reflecting strong diversification and balanced business structure.

# Recommendations

1. **Enhance Transaction Experience**
   Improve checkout efficiency, digital systems, and service processes in branches with rating gaps.

2. **Strengthen Core Regions**
   Sustain dominance in high-performing provinces through optimized inventory and targeted promotions.

3. **Stimulate Underperforming Areas**
   Refine regional marketing and operational strategies to increase profitability outside Java.

4. **Maintain Growth Momentum**
   Implement data-driven promotions and loyalty strategies to prevent revenue stagnation.

# Thank You

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